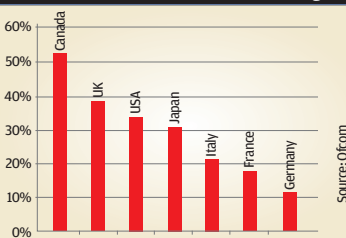


STAT ATTACK

Percentage of adults with Internet access who use social networking sites



Mobile subscriptions in 2007 touch 3.3 billion—about 50% of the world's population

Cell phone users in China are the biggest users of text messaging services, with an average of 967 messages per person per year.

IBM has 73,000 employees in India, a 40% increase from 2006.

UK gamers spent £1.52 bn on games (Rs 12,300 crore) in 2007. 78 million game titles were sold; games accounted for 79% of all software

competition is, in general, good, and this proliferation of free content will only benefit the end user!

DATA RAYS

Playing With Photons

With each generation, processors are getting smaller and more powerful, which you know all too well; the current emphasis is on multiple cores, higher bus speeds, and larger caches. However, some basic principles of

processor design have remained unchanged over the years. One of these is the mode of data transfer—wires have always been used for transferring data within a single core and also between cores.

There are problems, however. At the basic level, transfer of data means the flow of electrons, and this produces heat—which is a huge problem when it comes to modern processors. Electrons also can't travel faster than a certain velocity. The second factor has little bearing on quad-core systems, but when the number of cores becomes something like 12

(Intel and AMD already have plans for octa-core processors in two to three years), this architecture will be inadequate. Companies like IBM and Intel have been pouring millions of dollars into research on new processor fabrication designs and allied technologies; one result of these efforts is IBM's advances in transferring data at the core level using light.

IBM's researchers have improved upon the Mach-Zehnder electro-optic modulator, which is, at 200 micrometers long, more than 100 times smaller than similar devices. This modulator converts electric signals into light pulses. A laser beam is kept focused on the modulator, and when the electric signal carrying data falls on the modulator, it gets converted into light. This light beam falls on the other core's modulator, where an inverse process converts light to electronic signals. Since light travels faster than electrons in a wire, this means that data flow would be faster, resulting in the possibility of higher clock speeds. Since there is no heat production when light flows, this increased performance won't come with higher heat emissions, which is the bane of today's processors. Since cores are separated by 200 to 400 micrometers, this optical modulator seems possible in theory.

In practice, researchers say there is a lot of work to be done before chips using

this technology reach store shelves. Size is a big factor to consider; it is very difficult to manufacture optical components at such micro-scales. Besides, the processor architecture needs to be changed if this is to be implemented.

DEEP WATERS

Standard Trouble

Microsoft has several times been accused of unfair business practices, and has been rapped both by US and European free trade regulators for throttling competition. All these years, the company has been able to shrug off the accusations owing to the strong performance of its products. However, with the lower-than-expected performance of Vista and Google breathing down its neck, it looks like they've been on a sticky wicket and are open to mending their ways...

Pursuant to this new "strategy," Microsoft recently announced it would sign a deal with the makers of open source server software Samba, which will enable Samba software (which allows Linux users to share file and print services on Windows machines) to run smoothly on Windows. This deal was the result of a 2004 EU ruling, backed by another one in September, which held the software manufacturer guilty of not sharing

Android

Speculation has been rife in tech circles about an upcoming "Gphone," from Google. They went an entirely different way though, releasing a software stack for mobile phones, called Android.

Android is a combination of operating system, user interface, and Java-based applications like e-mail client, SMS, browser, maps, contacts, database management, etc. Built on the Linux kernel 2.6, Android development is open source and platform-independent. Essentially, this means that software that

is developed will run on a wide variety of handsets.

The Open Handset Alliance, a conglomeration of 34 companies—handset manufacturers (Motorola, LG), carriers (T-Mobile, Sprint), hardware companies (Intel, NVIDIA, Qualcomm), and software companies (eBay, Google)—is supporting the development of Android.

Google has released the Android SDK for developers, and has earmarked \$10 million as prize money for innovative applications. Phones working on Android are expected to be shipped by the end of 2008.

Buzzword of the MONTH